Polymers

Learning outcomes

By the end of this section you should be able to:

- Describe what polymers are.
- Describe the common properties of plastics in terms of their structure.

In our discussion of ionic, covalent, metallic bonding we have encountered several different solids. We have looked at ionic minerals, ionic salts, molecular solids, covalent network structures, metals and alloys. There is one type of solid material that is very commonly used globally that we have not mentioned so far. Can you think of what that might be? Do you need a hint? Look around the room you are in now. What substances are objects made of? Metals, wood, glass...

Another common type of material that is used to make a very wide range of objects are polymers. Usually, plastics are the first thing to come to mind when people hear the word polymers, but plastics are not the only type of polymers, as you will see throughout this and the following sections in the subtopic.

But what are polymers? What type of atoms create polymers? What is the structure and bonding of a polymer? Why does it have properties that societies find so useful? Why don't they all have the same properties? Why are some of these polymers a global environmental concern?

The term polymer is an Ancient Greek word meaning 'many parts'. Polymers are large molecules, or macromolecules, made from repeating subunits called monomers. These monomers are small molecules that make up the building blocks of a polymer. **Figure 1** below uses building blocks to illustrate the concept of a polymer having many parts.

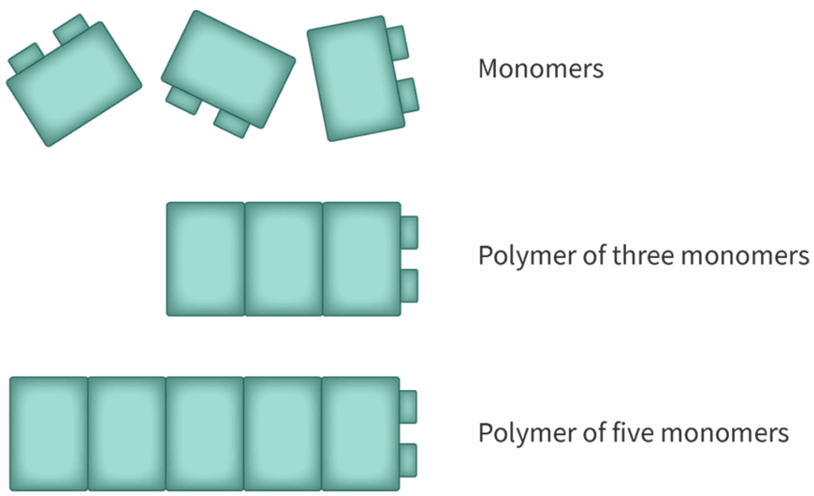

Figure 1. Polymers are macromolecules made up of monomers as the building blocks.

More information for figure 1

The image illustrates the structure of a polymer using a block model, conveying the concept of macromolecules composed of repeating subunits called monomers. The polymer is represented by multiple small block-like structures arranged in sequences to form a larger assembly, visually likening each block to a monomer within the polymer chain. The blocks are oriented in a grid-like fashion, showcasing multiple rows of connected blocks to emphasize their repetitive nature. This arrangement symbolizes how polymers are made from repeating units that build complex macromolecules.

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Recall from section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) that carbon can form a variety of different covalent network structures, called allotropes in addition to the large variety of small molecules. This is largely due in part to the four valence electrons allowing the formation of four covalent bonds. Carbon is also the backbone of most polymer structures. **Figure 2** shows an example of a polymer called polypropene (also known as polypropylene), made up only of carbon and hydrogen. Notice how it is a large molecule, hence, macromolecule, with many parts.

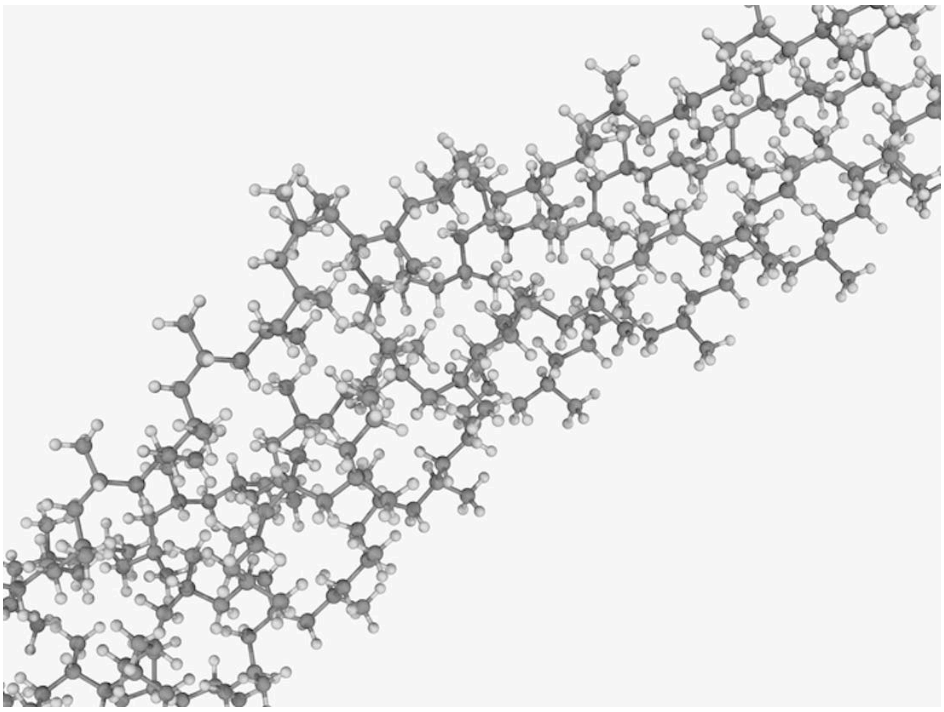


Figure 2. Example of a polymer structure, polypropene.

Credit: LAGUNA DESIGN, Getty Images

More information for figure 2

The image illustrates the molecular structure of polypropene, a synthetic polymer composed of repeating units derived from propylene monomers. The structure shows multiple chains of carbon and hydrogen atoms interconnected to form a larger macromolecule. This visual gives a detailed account of how the individual carbon atoms bond together, showcasing its complexity and the repeating pattern inherent in polymers. The structure is three-dimensional and complex, with each unit connecting to others in a network-like configuration.

[Generated by AI]

There are other ways that we can represent the polymer polypropene. In **Figure 2** we can see that it is a large molecule with many parts, but it is more difficult to see what the repeating unit actually is. Structural formulas are one way to show how atoms are bonded in the polymer structure that clearly shows the repeating unit. **Figure 3** shows a few different ways that the structural formula can be presented.

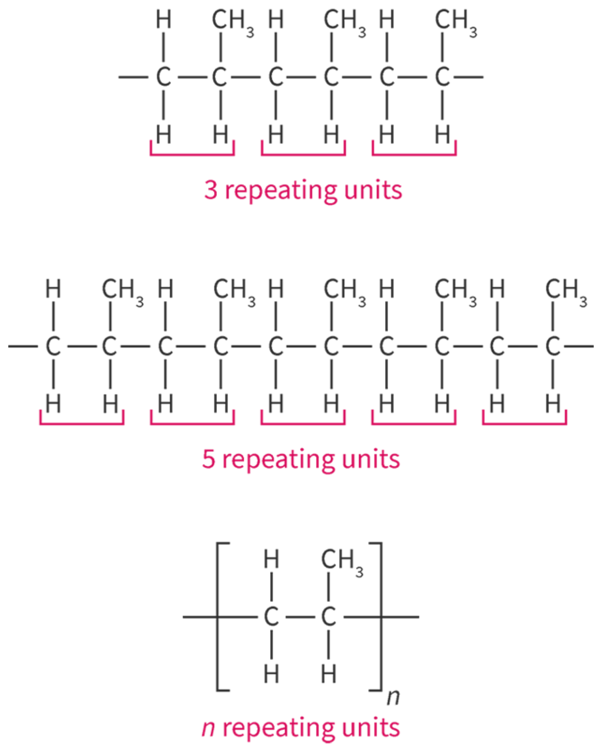


Figure 3. Structural formulas for polypropene.

More information for figure 3

The image illustrates different structural formulas of polypropene. It shows three variations: the first with three repeating units, the second with five, and the third with "n" repeating units, indicating continuation. Each formula is depicted with covalent bonds at the ends of the polymer chains left incomplete, suggesting the polymer's potential to extend indefinitely in either direction. This visualization helps in identifying the repeating unit of the polymer and understanding the polymer structure more clearly.

[Generated by AI]

Notice from **Figure 3** how the structural formula makes it much easier to identify the repeating unit of the polymer. The first example shows three repeating units, the second one shows five repeating units, and the third one shows *n* repeating units. The variable *n* is often used to denote that there can be any number of repeating units, 100, 1 million and usually even more! You will notice that at the end of the polymer chains the covalent bonds drawn are left incomplete to indicate the polymer can continue to repeat in either direction.

Plastic

Plastic is a category of polymer that can be moulded or extruded into a desired shape. Plastics are a wide range of materials used for consumer products, medical products, packaging, textiles, electronics, transportation, construction and more. **Figure 4** shows some common examples of plastic polymers and their structural formulas.

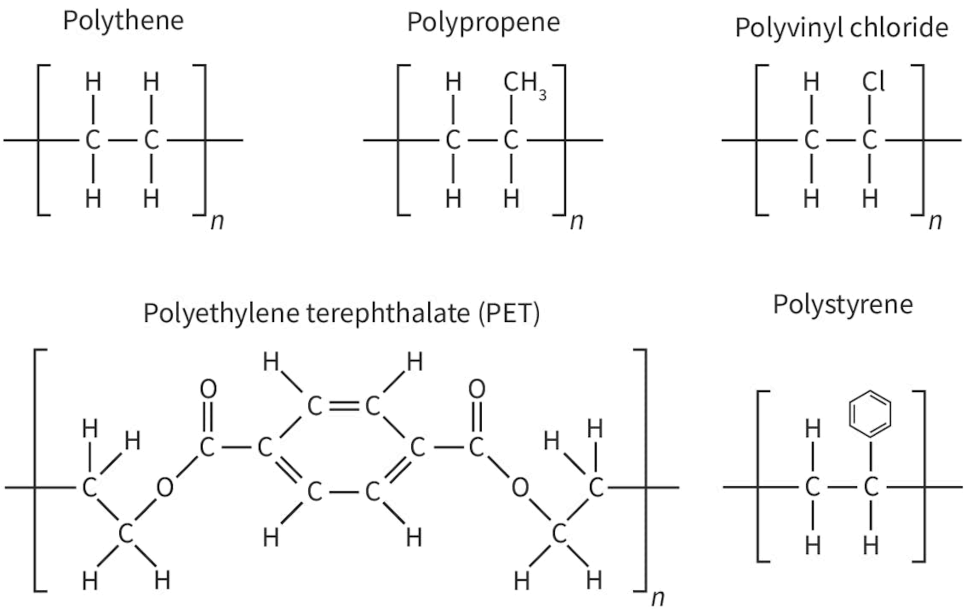


Figure 4. Common plastic polymers.

More information for figure 4

The image displays the structural formulas of six common plastic polymers: polythene, polypropene, polyvinyl chloride, polyethylene terephthalate (PET), and polystyrene.

1. **Polythene:** Shows a repeated unit of two carbon atoms (C) each bonded to two hydrogen atoms (H), with linear bonds extending to the sides indicating continuation.

2. **Polypropene:** Similar to polythene, but one hydrogen in the repeated unit is replaced with a methyl group (CH_3) on the second carbon.

3. **Polyvinyl chloride:** Formed by a repeated unit of two carbon atoms where one hydrogen is replaced by a chlorine atom (Cl) on the second carbon.

4. **Polyethylene terephthalate (PET):** Displays a complex chain structure with ester linkages (—COO—) and a phenyl ring in the middle.

5. **Polystyrene:** Consists of a repeated unit of two carbon atoms, with one hydrogen replaced by a phenyl group (a benzene ring with alternating double bonds) on the second carbon.

Each polymer is labeled above its structural formula. The repeated units are enclosed in square brackets with a subscript 'n' to indicate polymerization.

[Generated by AI]

Some of the main properties of plastics include:

- Flexible
- Durable
- Mould easily into desired shapes
- Lightweight
- Good electrical and thermal insulators

The polymer polypropene we looked at in **Figure 2** and **Figure 3** is also a plastic polymer. What do you think it is about its structure that allows it to have these properties? In subtopic S2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) you looked at molecular and covalent network structures. What was the large difference in structure that caused covalent network structures to be hard, rigid and have high melting points in comparison to molecular solids?

Covalent network solids formed three-dimensional lattice structures held together by covalent bonds whereas molecular solids involve individual atoms covalently bonded within molecules that are held together by intermolecular forces. Polymer structures are somewhere in between these two. They involve very long chains of molecules, but the individual polymer chains are held together by intermolecular forces. The intermolecular forces holding separate polymer chains together is similar to how intermolecular forces hold separate sheets together within graphite. A comparison of the structures of a molecular solids, covalent network and polymer are shown in **Interactive 1**. Click through each object to zoom in on the structure.

Interactive 1. Interactions Between Polypropene Polymer Chains.

More information for interactive 1

An interactive collage features three images of materials formed by covalent bonding. Each image has a plus sign, indicating an interactive hotspot. Selecting these hotspots reveals the description and chemical structure of the materials.
Left: Dry ice
Image: A white block of dry ice on a glass container, with white smoke coming out.
Structure: A cubic structure with CO_2 molecules occupying the edges and face centers of the cube.
Description on clicking the hotspot: Covalent bonds within each CO_2 molecule. Molecules held together by intermolecular forces in solid.
Middle: Polypropylene plastic bottle
Image: Three colored polypropylene plastic bottles displayed together.
Structure: The chemical structure features three rows of polypropylene. Polypropylene is made up of repeating propene (C_3H_6) units. Each row features a long carbon chain with alternating hydrogen (H) and methyl (CH_3) attached on the top and hydrogen (H) atoms attached to the bottom.
Description on clicking the hotspot: Covalent bonds within each polymer chain. Polymer chains held together by intermolecular forces.
Right: Diamond
Image: A shiny, reflective diamond.
Structure: A 3D network structure of carbon atoms arranged in four layers. Each carbon atom is tetrahedrally bonded to four other carbon atoms.
Description on clicking the hotspot: Held together by covalent bonds.

Flexibility and durability

You can see in **Interactive 1** that polymers have weak intermolecular forces that hold polymer chains together. These intermolecular forces are what gives plastic polymers flexibility and durability. Polymer chains can slide past one another but the polymer itself remains intact. Since flexibility is the ability of a material to be bent without breaking, the weaker the intermolecular forces between polymer chains, the more flexible a plastic is. Conversely, the stronger the intermolecular forces between polymer chains, the more rigid a plastic is.

Durability is the ability of a material to be functional over extended periods of time. Polymer chains are made up of a very large number of covalent bonds that do not break apart easily. This results in a long-lasting material that is durable. While the durability of plastics is useful can you think of why they might be of concern? Since the large number of covalent bonds do not break apart easily, they are difficult to break down. Once a plastic is made it is very difficult to get rid of. This is a global concern as plastics accumulate in landfills all over the world.

Nature of Science

Aspect: Global impact of science

Did you know there are newer plastics out there being developed to help reduce waste? These plastics are called bioplastics made up of starches, cellulose or biopolymers, obtained from renewable resources such as food waste. When these plastics are disposed of in the landfill, ordinary bacteria can break down the starch. Learn about the production, challenges and future of bioplastics in this article (https://www.sciencedirect.com/science/article/abs/pii/S2214799320301107).

Ability to mould into a desired shape

Since polymers are formed from smaller molecules that make up the repeating subunits called monomers, they can be moulded into shape during the formation of the polymer as monomers react together. In addition, since polymer chains are held together by weak intermolecular forces, many polymer plastics can be melted and reshaped to form new materials. You may have encountered this if you have placed something plastic too close to a heat source like a stove. A plastic container for example will melt and reshape when placed too close to the stove.

Lightweight

Considering the structure of polymers and the types of atoms that make up many plastic polymers, why do you think plastics are lightweight? Polymer chains are held together by weak intermolecular forces in polymers. Since this interaction is much weaker than covalent, metallic or ionic bonds, polymers do not pack very closely together in comparison to ionic, covalent networks and metallic lattices. This leads to polymers having a lower density than solids with an ionic, covalent network, or metallic lattice. Additionally, most polymer plastics are made up of long chains of carbon atoms with primarily hydrogen, oxygen, nitrogen and chlorine covalently bonded to these carbon chains. Carbon has a relatively low atomic mass, further contributing to the low density, lightweight nature of plastics.

Insulating

You learned about the electrical and thermal conductivity of metals in section S2.3.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/). Have you noticed how plastic is often used to coat electrical wires and the handles on cookware? If you look at the plastic polymer structures in **Figure 4**, can you predict why plastic polymers are poor electrical and thermal conductors? In all of the plastic polymer examples, there are no examples where electrons can delocalise throughout the polymer chains. This makes plastic polymers poor electrical and thermal conductors, which therefore make them good electrical and thermal insulators.

International Mindedness

The development of the polymer and plastics industry was actually motivated from trying to preserve natural resources.

1. The first synthetic polymer was produced by John Wesley Hyatt. He was looking to produce a material that could substitute the use of ivory in billiard balls. As the game billiards increased in popularity, natural ivory from elephant tusks became very scarce. The plastic produced by John Hyatt was developed to protect the natural world from human desires.
2. During World War II, natural resources became very scarce. New synthetic polymers were used to preserve natural resources and provide substitutes for military items.

The irony is now the production, consumption and disposal of plastic pose one of the world's largest threats. Responsible consumption and production is #12 of the UN's sustainable development goals (https://sdgs.un.org/goals). We all have a responsibility at the local, national and international level to evaluate the sustainability of production, reduce our consumption and re-evaluate the disposal of plastics.

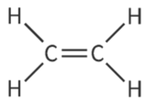
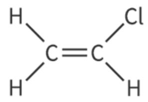
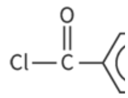
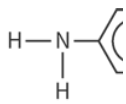
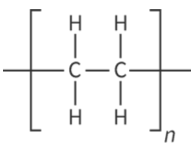
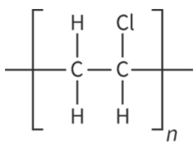
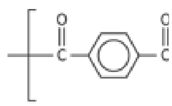
Properties of polymers and intermolecular interactions

The intermolecular interactions between polymer chains are correlated with the observed physical properties. Three commonly encountered plastic polymers are:

- Polyethylene (also known as polyethene)
- Polyvinylchloride (also known as PVC or polychloroethene)
- Kevlar

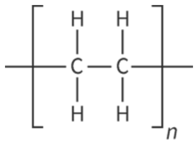
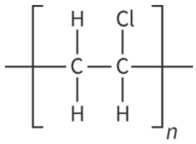
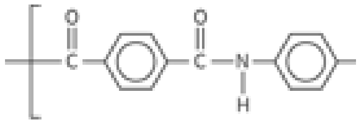
Table 1 shows the structures for the polymer and monomer for these three polymers. Look at each of them and identify what intermolecular forces (London dispersion forces, dipole-dipole forces, and hydrogen bonding) you would expect to exist between polymer chains. Which would you expect to be the strongest plastic polymer by looking at the structure of the monomers and polymers? Which would you expect to be the most flexible plastic?

1. Structural formula of three polymers and the monomers they are are pro

Polyethylene	Polyvinylchloride	Ke
 More information	 More information	 a 
 More information		 Mo

As these three polymers each have a different dominant intermolecular force between polymer chains, plastics made from these polymers will have differing properties. **Table 2** provides a summary of these intermolecular forces, properties and an example of where these polymers are used.

Table 2. Intermolecular forces and properties of three commonly encountered plastic poly

Polymer	Polyethylene	Polyvinylchloride	Kevlar
Polymer structure			
Intermolecular forces between polymer chains	London dispersion forces	London dispersion forces and dipole-dipole forces	London dispersion forces hydrogen bonding
Properties	Flexible	Rigid	Very strong and tough
Example of use	Plastic bags	Plumbing	Bulletproof vests

Natural polymers

So far, we have considered only synthetic polymers. Synthetic materials are those made by humans that cannot be found in nature. Monomers for synthetic polymers are usually derived from fossil fuels. Polymers have been around long before we humans have been synthesising them! Polymers found in nature are referred to as natural polymers. Here are some examples of natural polymers:

- Sugars such as cellulose found in wood and plants, and starch found in foods such as rice, potatoes and wheat.
- Proteins making up your skin, hair, muscles and nails are polymers made out of amino acids as the repeating units.
- DNA making up your genetic code is made up of two polymer strands that interact with one another through hydrogen bonding forming a double helix.
- Spider silk made from a polymer that has structural similarities to Kevlar.

Interactive 2 shows the structures of some of these natural polymers. Scroll through to see them.

Interactive 2. Examples of Natural Polymers.

More information for interactive 2

Making connections

Synthetic polymers are typically produced using starting materials from petroleum oil. The mass production of plastics globally has increased the demand on fossil fuels. To reduce the demand on fossil fuels but still produce plastic, a group of polymers known as biodegradable polymers has been created. Here are some examples of these structures.

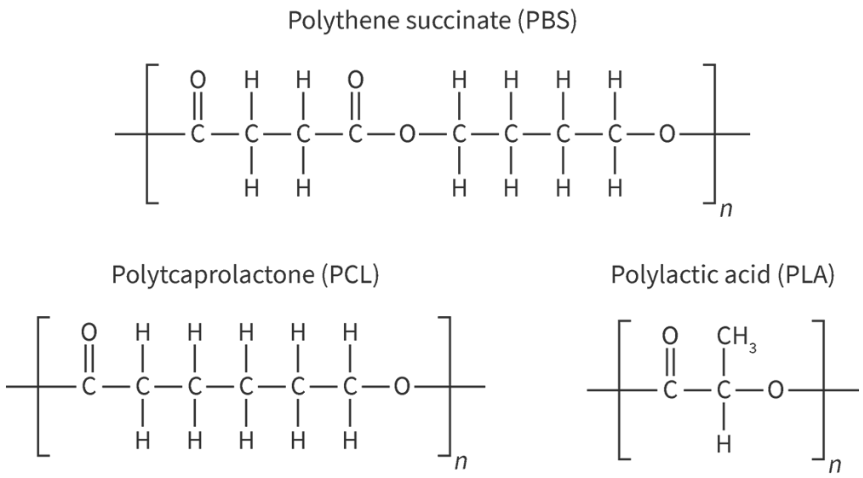


Figure 5. Examples of biodegradable polymers.

More information for figure 5

What structural features do these polymers have in common? Are they more structurally similar to synthetic or natural polymers? What structural features make them biodegradable?

Before completing this activity you will need to collect all the plastic waste you generate or encounter in a 24 hour time period. You will then work with a group to categorise these plastic polymers based on your physical observations.

Activity

- **IB learner profile attribute:** Caring
- **Approaches to learning:** Communication skills — Practising active listening skills
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group and whole class

Individually

Over the next 24 hours collect all of the plastic waste or products that you encounter and bring them to class. You should aim to bring in a minimum of five plastic objects.

In groups of 4—5

1. Gather all of your plastic objects and inspect them.
2. Create categories of the plastics using your physical observations of the similarities and differences of the objects. Come up with a title for each category.
3. Create a list of five advantages and five disadvantages of plastic polymers.
4. Research a local recycling program (could be from your school, municipal, district, etc.). Identify what types of plastics are recycled and if any information is provided about what plastic polymers are recycled.
5. Identify an alternative non-plastic option for each of the items each group member brought in.

As a class

1. In a class discussion, each group will share their categories for their plastic objects providing justification for how they decided upon the placement of their objects into the categories.
2. Create a class list of advantages and disadvantages of plastic polymers.
3. Discuss what are individual actions each of us can take to work towards meeting the UN's sustainable development goals (https://sdgs.un.org/goals) #12 responsible consumption and production.